

Advanced (Carolina Reaper)

Q1. Solve for x : $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Solve for y : $3^{y+1} = 27$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q3. Solve for z : $z^2 = 2^4$

- (A) $z = pm2$
- (B) $z = pm4$
- (C) $z = pm8$
- (D) $z = pm16$
- (E) $z = pm32$

Q4. If $a^x = b$ and $b^y = c$, what is a^{xy} ?

- (A) $a^{xy} = b$
- (B) $a^{xy} = c$
- (C) $a^{xy} = c^2$
- (D) $a^{xy} = b^2$
- (E) $a^{xy} = a^2$

Q5. Solve for x : $x^3 = 64$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q6. If $a^{2x} = 16$ and $a^x = 2$, what is the value of a ?

- (A) $a = 2$
- (B) $a = 4$
- (C) $a = 8$
- (D) $a = 16$
- (E) $a = 32$

Q7. Solve for y : $y^4 = 81$

- (A) $y = pm1$
- (B) $y = pm2$
- (C) $y = pm3$
- (D) $y = pm4$
- (E) $y = pm5$

Q8. Solve for z : $z^2 = 2^{2x}$

- (A) $z = 2^x$
- (B) $z = 2^{2x}$
- (C) $z = 4^x$
- (D) $z = 4^{2x}$
- (E) $z = 8^x$

Open Ended Questions (FRQ)

Q9. Solve the equation $x^{2y} = (x^y)^2$ for x and y . Give your final answer in terms of x and y .

Q10. Given that $a^{3x} = b$ and $a^{2y} = c$, find the value of x in terms of y and a , b , and c . Show all your work.

Chef's Tips

- Emphasize the importance of applying the properties of exponents when solving equations
- Remind students to check their solutions for extraneous solutions
- Encourage students to use graphs to visualize and check their solutions where necessary